



Mahindra University Hyderabad
École Centrale School of Engineering
Minor II

Program: B. Tech. Branch: CM Year: II Semester: Fall
 Subject: Real Analysis (MA2104)

Date: 30/10/2025
 Time Duration: 1.5 Hours

Start Time: 10.00 AM
 Max. Marks: 20

Instructions:

1. All questions are compulsory.
2. Please start each answer on a separate page.

Q 1:

4 marks

- (i) Define $f : (0, 1) \rightarrow \mathbb{R}$ by $f(x) = \frac{1}{x}$. Show that f is continuous on $(0, 1)$ but not uniformly continuous on $(0, 1)$. [2 marks]
- (ii) Give an example of a bounded function which is not Riemann integrable. Justify your answer. [2 marks]

Q 2:

3 marks

Show that every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Q 3:

3 marks

Let $(f_n) : [a, b] \rightarrow \mathbb{R}$ be a sequence of continuous functions and (f_n) converges uniformly to f on $[a, b]$. Show that f is continuous on $[a, b]$.

Q 4:

4 marks

Show that a bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann-Stieltjes integrable iff for each $\epsilon > 0$, there exists a partition $\mathbf{P}$ such that

$$U(f, \mathbf{P}, \alpha) - L(f, \mathbf{P}, \alpha) < \epsilon$$

where $\alpha : [a, b] \rightarrow \mathbb{R}$ is a monotonically increasing function.

Q 5:

6 marks

(i) Let $f(x) = e^x$ on $(0, \infty)$. Is f uniformly continuous on $(0, \infty)$. Justify your answer.

[2 marks]

(ii) Give an example of a functions f_n such that

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) dx \neq \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx.$$

Justify your answer.

[2 marks]

(iii) Find $\int_0^4 x^2 d[x]$.

[2 marks]

— End of Question Paper —